Meta-Regression

This vignette introduces the regression functionality of the `dtametaTMB` package for meta-analysis of diagnostic test accuracy (DTA) studies.

Validation

The subgroup HSROC and Reitsma implementations reproduce the Cochrane Handbook RF and anticcp examples closely, including the baseline parameters (HSROC: accuracy, threshold, shape // Reitsma: logit sensitivity and specificity), between-study variance estimates, and subgroup effects on accuracy and threshold. Small numerical differences may arise from optimizer settings, numerical tolerances, and implementation details, but all benchmark estimates and standard errors were recovered to close agreement.

Dummy-coding vs. cell-means parameterization

For subgroup analyses, we distinguish between two equivalent parameterizations of the same model. In the reference-group parameterization, one subgroup serves as the baseline and the remaining subgroup effects are expressed as deviations from this reference. In the group-specific (cell-means) parameterization, each subgroup has its own parameter directly. The former is convenient for testing subgroup differences, whereas the latter is convenient for reporting subgroup-specific estimates and plotting subgroup-specific HSROC curves. The Reitsma subgroup model is fitted twice with both parameterizations. The subgroup HSROC model is estimated using a reference-group parameterization, and subgroup-specific parameters are recovered by evaluating the fitted linear predictors at subgroup-specific design points via Z_{pred}.

How do we include covariates in the Reitsma model?

For sensitivity in study i, we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

Now we introduce a p-dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\begin{pmatrix} z_i^\top \mu_{Ai} \\ z_i^\top \mu_{Bi} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} z_i^\top \mu_A \\ z_i^\top \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

Let's assume that z_i includes a single binary covariate representing two subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \mu_{Ai} = \begin{cases} \mu_{Ai} & \text{for subgroup 1,} \\ \mu_{Ai} + \nu_{A2} & \text{for subgroup 2,} \end{cases} \qquad z_i^\top \mu_{Bi} = \begin{cases} \mu_{Bi} & \text{for subgroup 1,} \\ \mu_{Bi} + \nu_{B2} & \text{for subgroup 2,} \end{cases}$$

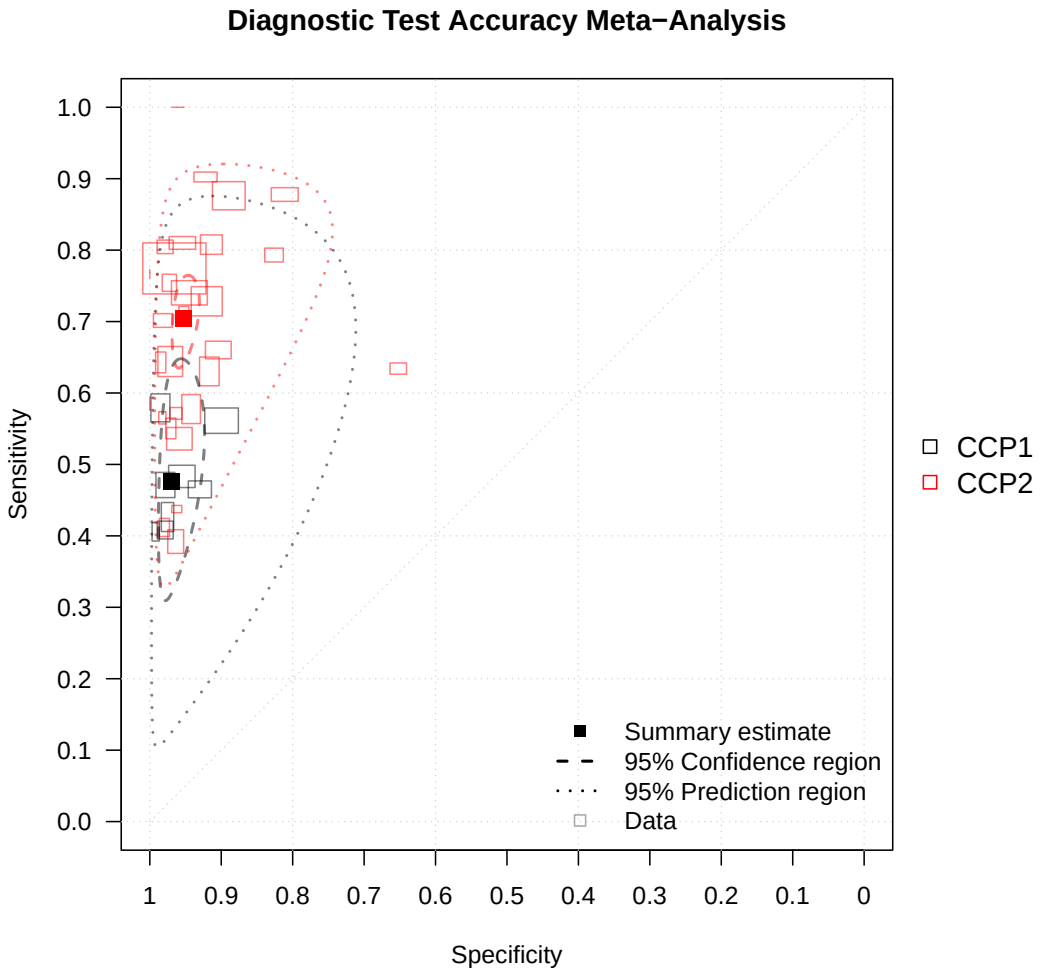
$$z_i^\top \mu_A = \begin{cases} \mu_A & \text{for subgroup 1,} \\ \mu_A + \nu_{A2} & \text{for subgroup 2,} \end{cases} \qquad z_i^\top \mu_B = \begin{cases} \mu_B & \text{for subgroup 1,} \\ \mu_B + \nu_{B2} & \text{for subgroup 2.} \end{cases}$$

```
data("anticcp")
reitsmasub <- fitReitsmaSubgroup(data=anticcp,
                                 TP=TP,
                                 FP=FP,
                                 FN=FN,
                                 TN=TN,
                                 study=study,
                                 subgroup=generation)
reitsmasub
#>
#> Reitsma Subgroup Model
#> ----------------------
#>
#> Number of studies   : 37
#> Number of subgroups : 2
#> Model fit           : Converged
#> -2 log likelihood   : 533.37
#> AIC                 : 547.37
#> BIC                 : 558.646
#>
#>
#> Use summary() for parameter estimates.
summary(reitsmasub)
#> $estimates
```

```
#>                   Estimate  Std_Error
#> mu_A.CCP1     -0.09653883 0.22032058
#> mu_B.CCP1      3.44671920 0.29824368
#> mu_A.CCP2      0.86603855 0.12087681
#> mu_B.CCP2      3.01649066 0.16223753
#> sigma2_A.sens  0.35982866 0.10217649
#> sigma2_B.spec  0.53990927 0.18015721
#> sigma_AB      -0.19684497 0.09835341
#> nu_A.CCP2      0.96257151 0.25134200
#> nu_B.CCP2     -0.43020982 0.33771185
#>
#> $sensspec
#>           type      Orig conflevel  CI_Lower  CI_Upper
#> mu_A.CCP1 sens 0.4758840      0.95 0.3708997 0.5830439
#> mu_B.CCP1 spec 0.9691331      0.95 0.9459445 0.9825578
#> mu_A.CCP2 sens 0.7039207      0.95 0.6522909 0.7508130
#> mu_B.CCP2 spec 0.9533136      0.95 0.9369387 0.9655926
#>
#> $RutterGatsonis_recovered
#>        Lambda    Theta      beta sigma2_alpha sigma2_theta
#> CCP1 3.727490 1.950971 0.2028866    0.4878424    0.3188056
#> CCP2 4.121047 1.278029 0.2028866    0.4878424    0.3188056
#>
#> $subgroups
#> [1] "CCP1" "CCP2"
```

How do I get a summary plot of the Reitsma model?

```
plot(reitsmasub,
     scale=0.01,
     nudge_legend=-0.2,
     size="se",
     col=c("black","red"))
```

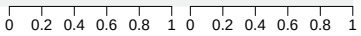


How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(reitsmasub,subgroup_label="Generation")
```

Study	Generation	TP	FP	FN	TN	Sensitivity (95%−CI)	Specificity (95%−CI)
Aotsuka 2005	CCP2	115	17	16	73	0.88 [0.81, 0.93]	0.81 [0.71, 0.89]
Bas 2003	CCP1	110	24	86	215	0.56 [0.49, 0.63]	0.90 [0.85, 0.93]
Bizzaro 2001	CCP1	40	5	58	227	0.41 [0.31, 0.51]	0.98 [0.95, 0.99]
Bombardieri 2004	CCP2	23	0	7	39	0.77 [0.58, 0.90]	1.00 [0.91, 1.00]
Choi 2005	CCP2	236	20	88	231	0.73 [0.68, 0.78]	0.92 [0.88, 0.95]
Correa 2004	CCP2	74	11	8	130	0.90 [0.82, 0.96]	0.92 [0.86, 0.96]
De Rycke 2004	CCP2	89	4	29	142	0.75 [0.67, 0.83]	0.97 [0.93, 0.99]
Dubucquoi 2004	CCP2	90	2	50	129	0.64 [0.56, 0.72]	0.98 [0.95, 1.00]
Fernandez−Suarez 2005	CCP2	31	0	22	75	0.58 [0.44, 0.72]	1.00 [0.95, 1.00]
Garcia−Berrocal 2005	CCP2	69	8	18	38	0.79 [0.69, 0.87]	0.83 [0.69, 0.92]
Girelli 2004	CCP2	25	2	10	40	0.71 [0.54, 0.85]	0.95 [0.84, 0.99]
Goldbach−Mansky 2000	CCP1	43	1	63	120	0.41 [0.31, 0.51]	0.99 [0.95, 1.00]
Greiner 2005	CCP2	70	5	17	228	0.80 [0.71, 0.88]	0.98 [0.95, 0.99]
Grootenboer−Mignot 2004	CCP2	167	8	98	88	0.63 [0.57, 0.69]	0.92 [0.84, 0.96]
Hitchon 2004	CCP2	26	8	15	15	0.63 [0.47, 0.78]	0.65 [0.43, 0.84]
Jansen 2003	CCP1	110	3	148	118	0.43 [0.37, 0.49]	0.98 [0.93, 0.99]
Kamali 2005	CCP2	26	1	20	56	0.57 [0.41, 0.71]	0.98 [0.91, 1.00]
Kumagai 2004	CCP2	64	14	15	293	0.81 [0.71, 0.89]	0.95 [0.92, 0.97]
Kwok 2005	CCP2	71	2	58	66	0.55 [0.46, 0.64]	0.97 [0.90, 1.00]
Lee 2003	CCP2	68	14	35	132	0.66 [0.56, 0.75]	0.90 [0.84, 0.95]
Lopez−Hoyos 2004	CCP2	38	3	0	73	1.00 [0.91, 1.00]	0.96 [0.89, 0.99]
Nell 2005	CCP2	42	2	60	96	0.41 [0.32, 0.51]	0.98 [0.93, 1.00]
Nielen 2005	CCP2	149	7	109	114	0.58 [0.51, 0.64]	0.94 [0.88, 0.98]
Quinn 2006	CCP2	147	10	35	106	0.81 [0.74, 0.86]	0.91 [0.85, 0.96]
Rantapaa−Dahlqvist 2003	CCP2	47	7	20	375	0.70 [0.58, 0.81]	0.98 [0.96, 0.99]
Raza 2005	CCP2	24	3	18	79	0.57 [0.41, 0.72]	0.96 [0.90, 0.99]
Saraux 2003	CCP1	40	11	46	146	0.47 [0.36, 0.58]	0.93 [0.88, 0.96]
Sauerland 2005	CCP2	171	26	60	443	0.74 [0.68, 0.80]	0.94 [0.92, 0.96]
Schellekens 2000	CCP1	72	14	77	298	0.48 [0.40, 0.57]	0.96 [0.93, 0.98]
Soderlin 2004	CCP2	7	2	9	51	0.44 [0.20, 0.70]	0.96 [0.87, 1.00]
Suzuki 2003	CCP2	481	23	68	185	0.88 [0.85, 0.90]	0.89 [0.84, 0.93]
Vallbracht 2004	CCP2	190	12	105	408	0.64 [0.59, 0.70]	0.97 [0.95, 0.99]
van Gaalen 2005	CCP2	82	13	71	301	0.54 [0.45, 0.62]	0.96 [0.93, 0.98]
van Venrooij 2004	CCP2	865	79	252	2218	0.77 [0.75, 0.80]	0.97 [0.96, 0.97]
Vincent 2002	CCP1	139	7	101	464	0.58 [0.51, 0.64]	0.99 [0.97, 0.99]
Vittecoq 2004	CCP2	69	5	107	133	0.39 [0.32, 0.47]	0.96 [0.92, 0.99]
Zeng 2003	CCP1	90	7	101	313	0.47 [0.40, 0.54]	0.98 [0.96, 0.99]



How do I constrain parameters in the Reitsma model?

In sparse data one may wish to fix parameters of the random effects (variance-covariance matrix) at zero. This can be done via the `constrain` argument.

For example,

```
constrainA <- fitReitsmaSubgroup(data=anticcp,
                                 TP=TP,
                                 FP=FP,
                                 FN=FN,
                                 TN=TN,
                                 study=study,
```

```
                                subgroup=generation,
                                constrain="sigma2_A")
summary(constrainA)$estimates
#>                     Estimate  Std_Error
#> mu_A.CCP1     -5.439405e-02 0.05498530
#> mu_B.CCP1      3.446625e+00 0.29875482
#> mu_A.CCP2      9.179794e-01 0.03139172
#> mu_B.CCP2      2.996859e+00 0.16216233
#> sigma2_A.sens  4.930381e-32 0.00000000
#> sigma2_B.spec  5.396331e-01 0.17997640
#> sigma_AB       0.000000e+00 0.00000000
#> nu_A.CCP2      9.723735e-01 0.06331527
#> nu_B.CCP2     -4.497661e-01 0.33795382
```

fixes the logit sensitivity variance to zero. This also implies that the random effects covariance is zero. Note that you can also set `constrain` to `"sigma2_AB"`, `"sigma2_B"`, or `"all"`.

Constraining fixed effects is controlled by the `subgroup_constrain` argument. For example,

```
constrainsens <- fitReitsmaSubgroup(data=anticcp,
                                    TP=TP,
                                    FP=FP,
                                    FN=FN,
                                    TN=TN,
                                    study=study,
                                    subgroup=generation,
                                    subgroup_constrain="sens")
summary(constrainsens)$estimates
#>                   Estimate Std_Error
#> mu_A.CCP1      0.65330520 0.1274076
#> mu_B.CCP1      3.08122030 0.3256178
#> mu_A.CCP2      0.65330520 0.0000000
#> mu_B.CCP2      3.11800628 0.1745128
#> sigma2_A.sens  0.54192926 0.1462735
#> sigma2_B.spec  0.57765267 0.1996838
#> sigma_AB      -0.27715832 0.1398143
#> nu_A.CCP2      0.00000000 0.0000000
#> nu_B.CCP2      0.03678907 0.3857733
```

assumes equal (logit) sensitivities in all subgroups. Note that you can also set `subgroup_constrain` to `"spec"` or `c("sens","spec")`.

How can I compare `constrainsens` with the full model?

We can perform a likelihood ratio test via `anova`, essentially testing whether there exist subgroup differences in sensitivity.

```
anova(constrainsens,reitsmasub)
#>         Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1  6 -272.77
#> Model 2  7 -266.69       1 12.181  0.0004827 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do we include covariates in the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

Now we introduce a p-dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\text{logit}(\pi_{ij}) = \left(z_i^\top \theta_i + z_i^\top \alpha_i x_{ij}\right) \exp(-z_i^\top \beta x_{ij}),$$

$$z_i^\top \alpha_i \sim \mathcal{N}(z_i^\top \Lambda, \sigma_\alpha^2), \qquad z_i^\top \theta_i \sim \mathcal{N}(z_i^\top \Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

Let's assume that z_i includes a single categorical covariate representing three subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \theta_i = \begin{cases} \theta_i & \text{for subgroup 1,} \\ \theta_i + \gamma_2 & \text{for subgroup 2,} \\ \theta_i + \gamma_3 & \text{for subgroup 3,} \end{cases} \qquad z_i^\top \alpha_i = \begin{cases} \alpha_i & \text{for subgroup 1,} \\ \alpha_i + \xi_2 & \text{for subgroup 2,} \\ \alpha_i + \xi_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \beta = \begin{cases} \beta & \text{for subgroup 1,} \\ \beta + \delta_2 & \text{for subgroup 2,} \\ \beta + \delta_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \Lambda = \begin{cases} \Lambda & \text{for subgroup 1,} \\ \Lambda + \xi_2 & \text{for subgroup 2,} \\ \Lambda + \xi_3 & \text{for subgroup 3,} \end{cases} \qquad z_i^\top \Theta_i = \begin{cases} \Theta & \text{for subgroup 1,} \\ \Theta + \gamma_2 & \text{for subgroup 2,} \\ \Theta + \gamma_3 & \text{for subgroup 3.} \end{cases}$$

Notes:

- By default, `fitRutterGatsonisSubgroup()` assumes a common shape across all subgroups, which implies $\delta_2 = \delta_3 = 0$ for the case above.
- Moreover, for prediction `fitRutterGatsonisSubgroup()` uses the prediction matrix

$$Z_{\text{pred}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

  for the case above and therefore recovers the threshold, accuracy, and shape parameters for each subgroup as if it were the reference group.

How do I perform a Rutter and Gatsonis meta-regression with a single categorical study-level covariate?

```
data("RF")
RF2        <- RF[RF$method %in% c("LA","ELISA","Nephelometry"),]
RF2$method <- factor(RF2$method,levels=c("LA","ELISA","Nephelometry"))
ruttergatsonissub <- fitRutterGatsonisSubgroup(data=RF2,
                                               TP=TP,
                                               FP=FP,
                                               FN=FN,
                                               TN=TN,
                                               study=study,
                                               subgroup=method,
                                               constrain="shape") # assumes equal
                                                                  # shapes in subgroups
```

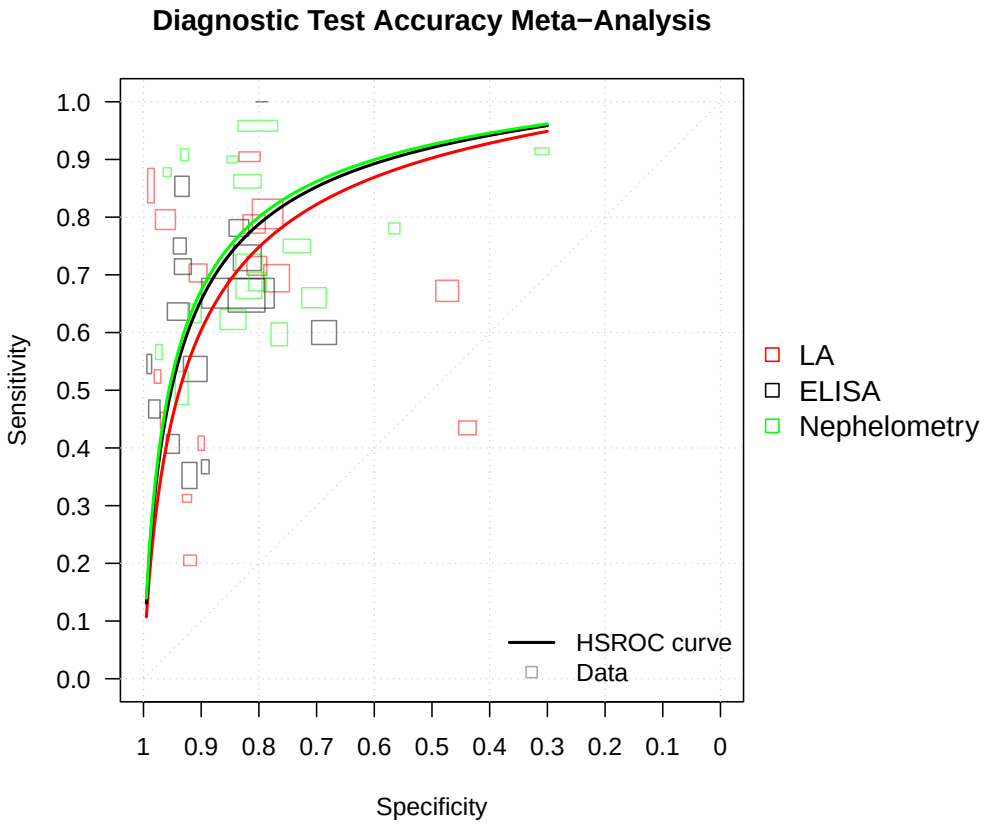
```
ruttergatsonissub
#>
#> Rutter & Gatsonis Subgroup Model
#> --------------------------------
#>
#> Number of studies   : 47
#> Number of subgroups : 3
```

```
#> Model fit           : Converged
#> -2 log likelihood   : 753.722
#> AIC                 : 771.722
#> BIC                 : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissub)
#> $estimates
#>                         Estimate Std. Error
#> Lambda_LA            2.45630704 0.32357707
#> xi_ELISA             0.24853054 0.43954126
#> xi_Nephelometry      0.33141751 0.44260069
#> Theta_LA            -0.55005213 0.21334057
#> gamma_ELISA         -0.19625909 0.26096239
#> gamma_Nephelometry   0.49586320 0.26223094
#> beta_LA              0.19768314 0.16983954
#> delta_ELISA          0.00000000 0.00000000
#> delta_Nephelometry   0.00000000 0.00000000
#> sigma2_alpha         1.27791582 0.30852514
#> sigma2_theta         0.47684216 0.11344574
#> Lambda_LA            2.45630704 0.32357707
#> Lambda_ELISA         2.70483758 0.32695434
#> Lambda_Nephelometry  2.78772454 0.30577914
#> Theta_LA            -0.55005213 0.21334057
#> Theta_ELISA         -0.74631122 0.20991330
#> Theta_Nephelometry  -0.05418893 0.21213149
#> beta_LA              0.19768314 0.16983954
#> beta_ELISA           0.19768314 0.16983954
#> beta_Nephelometry    0.19768314 0.16983954
#> logitsens            0.82616754 0.28629967
#> logitsens            1.05130793 0.28606921
#> logitsens            1.12639409 0.27689507
#> sens                 0.69554397 0.06062755
#> sens                 0.74102598 0.05489854
#> sens                 0.75517283 0.05119425
#>
#> $sensspec
#>       subgroup      spec conflevel logitsens Std_Error  CI_Lower CI_Upper
#> 1           LA 0.8461538      0.95 0.8261675 0.2862997 0.2650305 1.387305
#> 2        ELISA 0.8461538      0.95 1.0513079 0.2860692 0.4906226 1.611993
#> 3 Nephelometry 0.8461538      0.95 1.1263941 0.2768951 0.5836897 1.669098
```

```
#>        Sens SensCI_Lower SensCI_Upper
#> 1 0.6955440    0.5658725    0.8001616
#> 2 0.7410260    0.6202531    0.8336879
#> 3 0.7551728    0.6419160    0.8414556
#>
#> $Reitsma_recovered
#>              mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec   sigma_AB
#> LA           0.6142827  1.962946     0.6534849      0.970378 -0.1573632
#> ELISA        0.5490645  2.316770     0.6534849      0.970378 -0.1573632
#> Nephelometry 1.2135916  1.598491     0.6534849      0.970378 -0.1573632
#>
#> $subgroups
#> [1] "LA"           "ELISA"        "Nephelometry"
```

How do I get a summary plot of the Rutter and Gatsonis subgroup model?

```
plot(ruttergatsonissub,
     specrange=c(0.3,0.995),
     size="se",
     col=c("red","black","green"),
     scale=0.015)
```

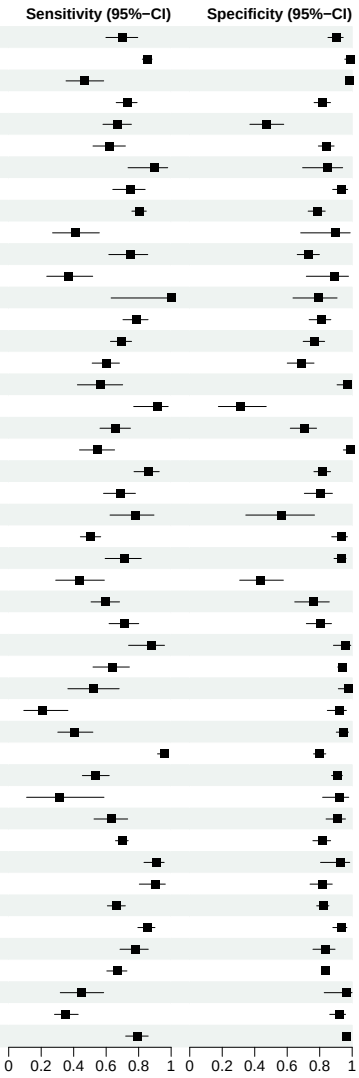


How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(ruttergatsonissub,subgroup_label = "Method")
```

Study	Method	TP	FP	FN	TN	Sensitivity (95%−CI)	Specificity (95%−CI)
Aho 1999	LA	64	16	27	153	0.70 [0.60, 0.79]	0.91 [0.85, 0.94]
Anuradha 2005	LA	482	2	82	153	0.85 [0.82, 0.88]	0.99 [0.95, 1.00]
Banchuin 1992	ELISA	36	6	41	313	0.47 [0.35, 0.58]	0.98 [0.96, 0.99]
Bas 2003	ELISA	143	43	53	196	0.73 [0.66, 0.79]	0.82 [0.77, 0.87]
Berthelot 1995	LA	80	50	39	45	0.67 [0.58, 0.76]	0.47 [0.37, 0.58]
Bizzaro 2001	Nephelometry	61	36	37	196	0.62 [0.52, 0.72]	0.84 [0.79, 0.89]
Bombardieri 2004	Nephelometry	27	6	3	33	0.90 [0.73, 0.98]	0.85 [0.69, 0.94]
Carpenter 1989	ELISA	60	8	20	119	0.75 [0.64, 0.84]	0.94 [0.88, 0.97]
Choi 2005	LA	261	54	63	197	0.81 [0.76, 0.85]	0.78 [0.73, 0.83]
Cordonnier 1996	LA	20	2	29	18	0.41 [0.27, 0.56]	0.90 [0.68, 0.99]
Das 2004	Nephelometry	42	46	14	127	0.75 [0.62, 0.86]	0.73 [0.66, 0.80]
Davis 1989	ELISA	18	3	31	25	0.37 [0.23, 0.52]	0.89 [0.72, 0.98]
de Bois 1996	ELISA	8	8	0	31	1.00 [0.63, 1.00]	0.79 [0.64, 0.91]
De Rycke 2004	LA	93	28	25	118	0.79 [0.70, 0.86]	0.81 [0.73, 0.87]
Despres 1994	LA	143	39	63	130	0.69 [0.63, 0.76]	0.77 [0.70, 0.83]
Dubucquoi 2004	ELISA	84	41	56	90	0.60 [0.51, 0.68]	0.69 [0.60, 0.77]
Fernandez−Suarez 2005	Nephelometry	30	2	23	73	0.57 [0.42, 0.70]	0.97 [0.91, 1.00]
Girelli 2004	Nephelometry	32	29	3	13	0.91 [0.77, 0.98]	0.31 [0.18, 0.47]
Goldbach−Mansky 2000	Nephelometry	70	39	36	93	0.66 [0.56, 0.75]	0.70 [0.62, 0.78]
Gomes−Daudrix 1994	ELISA	48	1	40	99	0.55 [0.44, 0.65]	0.99 [0.95, 1.00]
Greiner 2005	Nephelometry	75	42	12	191	0.86 [0.77, 0.93]	0.82 [0.76, 0.87]
Grootenboer−Mignot 2004	Nephelometry	64	18	29	73	0.69 [0.58, 0.78]	0.80 [0.71, 0.88]
Hitchon 2004	Nephelometry	32	10	9	13	0.78 [0.62, 0.89]	0.57 [0.34, 0.77]
Jansen 2003	Nephelometry	130	8	128	113	0.50 [0.44, 0.57]	0.93 [0.87, 0.97]
Jonsson 1998	ELISA	50	14	20	191	0.71 [0.59, 0.82]	0.93 [0.89, 0.96]
Kamali 2005	LA	20	32	26	25	0.43 [0.29, 0.59]	0.44 [0.31, 0.58]
Kwok 2005	Nephelometry	77	16	52	52	0.60 [0.51, 0.68]	0.76 [0.65, 0.86]
Lee 2003	LA	73	22	29	90	0.72 [0.62, 0.80]	0.80 [0.72, 0.87]
Lopez−Hoyos 2004	Nephelometry	36	3	5	70	0.88 [0.74, 0.96]	0.96 [0.88, 0.99]
Rantapaa−Dahlqvist 2003	ELISA	49	23	28	359	0.64 [0.52, 0.74]	0.94 [0.91, 0.96]
Raza 2005	LA	22	2	20	80	0.52 [0.36, 0.68]	0.98 [0.91, 1.00]
Saraux 1995	LA	8	8	31	91	0.21 [0.09, 0.36]	0.92 [0.85, 0.96]
Saraux 2003	ELISA	35	8	51	149	0.41 [0.30, 0.52]	0.95 [0.90, 0.98]
Sauerland 2005	Nephelometry	161	89	7	360	0.96 [0.92, 0.98]	0.80 [0.76, 0.84]
Schellekens 2000	ELISA	80	28	69	284	0.54 [0.45, 0.62]	0.91 [0.87, 0.94]
Soderlin 2004	LA	5	4	11	49	0.31 [0.11, 0.59]	0.92 [0.82, 0.98]
Spiritus 2004	Nephelometry	57	9	33	93	0.63 [0.53, 0.73]	0.91 [0.84, 0.96]
Suzuki 2003	Nephelometry	383	38	166	170	0.70 [0.66, 0.74]	0.82 [0.76, 0.87]
Swedler	Nephelometry	89	3	9	39	0.91 [0.83, 0.96]	0.93 [0.81, 0.99]
Thammanichanond 2005	LA	57	25	6	111	0.90 [0.80, 0.96]	0.82 [0.74, 0.88]
Vallbracht 2004	ELISA	196	75	99	345	0.66 [0.61, 0.72]	0.82 [0.78, 0.86]
van Leeuwen 1988	ELISA	163	10	28	140	0.85 [0.80, 0.90]	0.93 [0.88, 0.97]
Vasiliauskiene 2001	ELISA	75	21	21	106	0.78 [0.69, 0.86]	0.83 [0.76, 0.89]
Visser 1996	ELISA	157	287	78	1466	0.67 [0.60, 0.73]	0.84 [0.82, 0.85]
Vittecoq 2001	LA	26	1	32	29	0.45 [0.32, 0.58]	0.97 [0.83, 1.00]
Vittecoq 2004	ELISA	62	11	114	127	0.35 [0.28, 0.43]	0.92 [0.86, 0.96]
Winkles 1989	LA	113	19	29	481	0.80 [0.72, 0.86]	0.96 [0.94, 0.98]



How do I constrain parameters in the Rutter and Gatsonis subgroup model?

In the Rutter and Gatsonis model, all parameter constraints are controlled by `constrain`, i.e.,

- `constrain="sigma2_alpha"` sets σ^2_α to zero,
- `constrain="sigma2_theta"` sets σ^2_θ to zero,

- `constrain="accuracy"` assumes equal accuracy parameters across subgroups, i.e. $\xi_2 = \xi_3 = \cdots = 0$,
- `constrain="threshold"` assumes equal threshold parameters across subgroups, i.e. $\gamma_2 = \gamma_3 = \cdots = 0$,
- `constrain="shape"` assumes equal shape parameters across subgroups, i.e. $\delta_2 = \delta_3 = \cdots = 0$,
- `constrain="shape_zero"` fixes all shape parameters at zero.

Constraints can also be combined, for example `constrain=c("shape","sigma2_theta")`.

How do I use the general Rutter and Gatsonis regression function?

This method is for advanced users who feel comfortable specifying their own design and prediction matrices. For study-level covariates, the design matrix Z needs two identical consecutive rows per study, one for the diseased and one for the non-diseased. Of note, there are neither `plot()` nor `forest()` methods for `fitRutterGatsonisReg()`.

Let's reproduce the subgroup-analysis from before.

```
# Specify design matrix Z
Z  <- model.matrix(~method,data=RF2)
# For study level-covariates, we need to two identical consecutive
# rows per study (diseased and non-diseased).
Z2 <- Z[rep(seq_len(nrow(Z)), each = 2), , drop = FALSE]
# Specify prediction matrix Z_pred
Z_pred <- matrix(c(1,0,0,1,1,0,1,0,1),ncol=3,nrow=3,byrow=T)
constrain <- list(shape_coef=factor(c(1, rep(NA, ncol(Z2) - 1))))

ruttergatsonisreg <- fitRutterGatsonisReg(data=RF2,
                                          TP=TP,
                                          FP=FP,
                                          FN=FN,
                                          TN=TN,
                                          study=study,
                                          Z=Z2,
                                          Z_pred=Z_pred,
                                          map=constrain)

ruttergatsonisreg
#>
#> Rutter & Gatsonis Regression Model
#> ----------------------------------
#>
```

```
#> Number of studies : 47
#> Model fit         : Converged
#> -2 log likelihood : 753.722
#> AIC               : 771.722
#> BIC               : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonisreg)
#> $estimates
#>                   Estimate Std. Error
#> accuracy_coef   2.45631156 0.32357673
#> accuracy_coef   0.24852208 0.43954066
#> accuracy_coef   0.33141584 0.44260014
#> threshold_coef -0.55005084 0.21334149
#> threshold_coef -0.19625009 0.26096393
#> threshold_coef  0.49585695 0.26223232
#> shape_coef      0.19768191 0.16983950
#> shape_coef      0.00000000 0.00000000
#> shape_coef      0.00000000 0.00000000
#> sigma2_alpha    1.27791200 0.30852400
#> sigma2_theta    0.47684805 0.11344753
#> Lambda_Pred     2.45631156 0.32357673
#> Lambda_Pred     2.70483365 0.32695391
#> Lambda_Pred     2.78772741 0.30577879
#> Theta_Pred     -0.55005084 0.21334149
#> Theta_Pred     -0.74630093 0.20991415
#> Theta_Pred     -0.05419388 0.21213237
#> beta_Pred       0.19768191 0.16983950
#> beta_Pred       0.19768191 0.16983950
#> beta_Pred       0.19768191 0.16983950
#> logitsens       0.82617129 0.28629952
#> logitsens       1.05130416 0.28606887
#> logitsens       1.12639652 0.27689490
#> sens            0.69554476 0.06062743
#> sens            0.74102525 0.05489857
#> sens            0.75517328 0.05119416
#>
#> $sensspec
#>        spec conflevel logitsens Std_Error  CI_Lower CI_Upper      Sens
#> 1 0.8461538      0.95 0.8261713 0.2862995 0.2650345 1.387308 0.6955448
#> 2 0.8461538      0.95 1.0513042 0.2860689 0.4906195 1.611989 0.7410253
```

```
#> 3 0.8461538      0.95 1.1263965 0.2768949 0.5836925 1.669101 0.7551733
#>   SensCI_Lower SensCI_Upper
#> 1    0.5658735    0.8001621
#> 2    0.6202524    0.8336873
#> 3    0.6419166    0.8414559
```

How do I compare models?

In the previous section we fitted the `RF` data set using the Rutter and Gatsonis subgroup model while keeping the shape parameter equal across all subgroups `constrain="shape"`. Now let's fit the full model allowing for different shape parameters across subgroups and check whether the data lend support to this approach.

```
ruttergatsonissubfull <- fitRutterGatsonisSubgroup(data=RF2,
                                                   TP=TP,
                                                   FP=FP,
                                                   FN=FN,
                                                   TN=TN,
                                                   study=study,
                                                   subgroup=method,
                                                   constrain=NULL)
ruttergatsonissubfull
#>
#> Rutter & Gatsonis Subgroup Model
#> --------------------------------
#>
#> Number of studies   : 47
#> Number of subgroups : 3
#> Model fit           : Converged
#> -2 log likelihood   : 753.553
#> AIC                 : 775.553
#> BIC                 : 795.905
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissubfull)
#> $estimates
#>                       Estimate Std. Error
#> Lambda_LA            2.4243627 0.33049053
#> xi_ELISA             0.2974489 0.51315108
#> xi_Nephelometry      0.3716072 0.45086464
```

```
#> Theta_LA            -0.5029491 0.24451535
#> gamma_ELISA         -0.2578309 0.37024437
#> gamma_Nephelometry   0.3970582 0.35944143
#> beta_LA              0.2777423 0.26642216
#> delta_ELISA         -0.1028861 0.42660921
#> delta_Nephelometry  -0.1603851 0.39753663
#> sigma2_alpha         1.2730509 0.30798445
#> sigma2_theta         0.4762342 0.11343407
#> Lambda_LA            2.4243627 0.33049053
#> Lambda_ELISA         2.7218116 0.39299371
#> Lambda_Nephelometry  2.7959699 0.30708020
#> Theta_LA            -0.5029491 0.24451535
#> Theta_ELISA         -0.7607799 0.27816983
#> Theta_Nephelometry  -0.1058909 0.26322751
#> beta_LA              0.2777423 0.26642216
#> beta_ELISA           0.1748562 0.33321260
#> beta_Nephelometry    0.1173572 0.29500266
#> logitsens            0.8186872 0.27519077
#> logitsens            1.0626859 0.32405595
#> logitsens            1.1206500 0.28834948
#> sens                 0.6939576 0.05844514
#> sens                 0.7432035 0.06184675
#> sens                 0.7541093 0.05346821
#>
#> $sensspec
#>       subgroup      spec conflevel logitsens Std_Error  CI_Lower CI_Upper
#> 1           LA 0.8461538      0.95 0.8186872 0.2751908 0.2793232 1.358051
#> 2        ELISA 0.8461538      0.95 1.0626859 0.3240560 0.4275479 1.697824
#> 3 Nephelometry 0.8461538      0.95 1.1206500 0.2883495 0.5554954 1.685805
#>        Sens SensCI_Lower SensCI_Upper
#> 1 0.6939576    0.5693803    0.7954428
#> 2 0.7432035    0.6052880    0.8452503
#> 3 0.7541093    0.6354096    0.8436716
#>
#> $Reitsma_recovered
#>              mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec   sigma_AB
#> LA           0.6172733  1.970644     0.6018251     1.0488519 -0.1579715
#> ELISA        0.5498862  2.315531     0.6670420     0.9463054 -0.1579715
#> Nephelometry 1.2184573  1.594762     0.7065203     0.8934286 -0.1579715
#>
#> $subgroups
#> [1] "LA"           "ELISA"        "Nephelometry"
```

How do I get the log likelihood, the AIC, and the BIC of a model?

```
logLik(ruttergatsonissubfull)
#> 'log Lik.' -376.7765 (df=11)
AIC(ruttergatsonissubfull)
#> [1] 775.5531
BIC(ruttergatsonissubfull)
#> [1] 795.9047

logLik(ruttergatsonissub)
#> 'log Lik.' -376.8612 (df=9)
AIC(ruttergatsonissub)
#> [1] 771.7223
BIC(ruttergatsonissub)
#> [1] 788.3736
```

How do I perform a likelihood ratio test of the two models?

The test below formally investigates whether the HSROC curve shapes are equal in all subgroups.

```
anova(ruttergatsonissub,
      ruttergatsonissubfull)
#>         Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1  9 -376.86
#> Model 2 11 -376.78       2 0.1692     0.9189
```

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